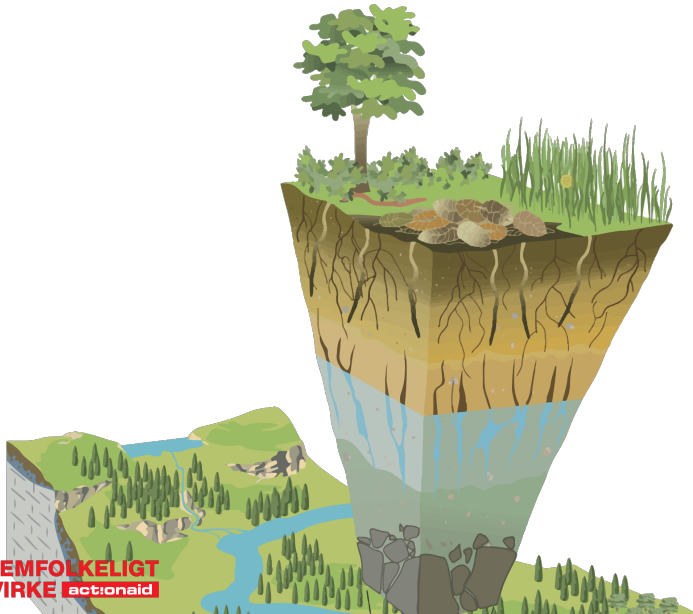


Don't Shit where You Drill!



Don't Shit where You Drill!

Michel Hardenberg

PhD student at AU - Quantifying the Vadose Zone
Department of Electrical and Computer Engineering

**MELLEMFOLKELIGT
SAMVIRKE** **act:onaid**

CLIMATE JUSTICE SOCIETY SEPTEMBER 19, 2026

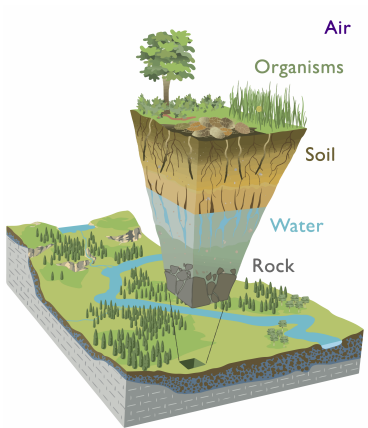
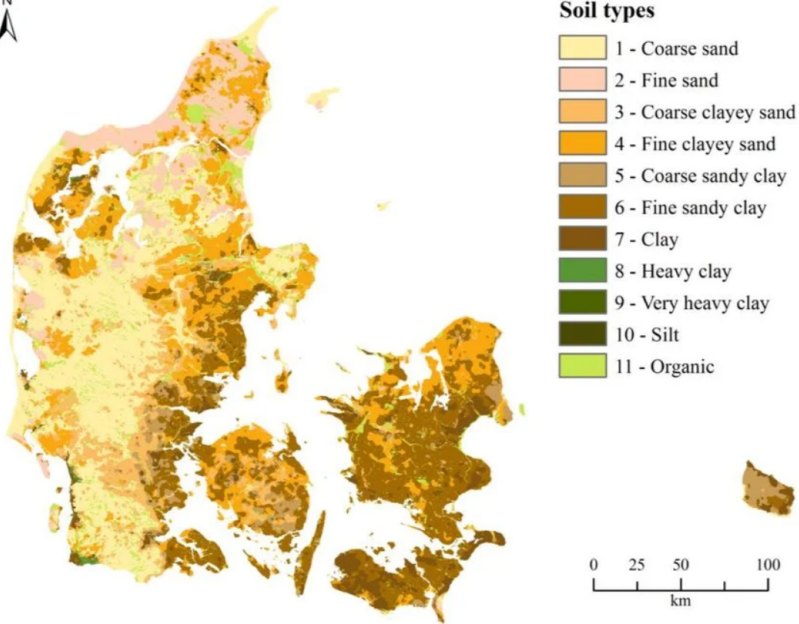


Illustration of soil layers in the near surface [Chorover et al., 2007].

APSU - magnetic measurements



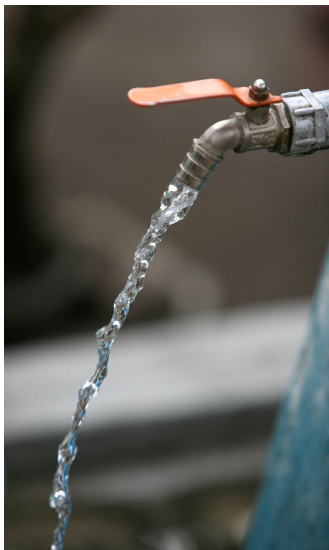
TTEM- electrical measurements



ToC

Part One:

- ▶ What are the culprits?
- ▶ How are we looking?
- ▶ What is the government doing?
Promising?



GEUS WATER REPORT 2025



¹PFOS, PFOA, PFBS, PFHxS

GEUS WATER REPORT 2025

Pesticides are found in 70% of testcites. 1 of 3 exceed the safety limits.



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Nitrate from excessive farming reaches high concentrations in **1 of 3** sites. 16% break the limits. In shallow depth sites, 31% do.



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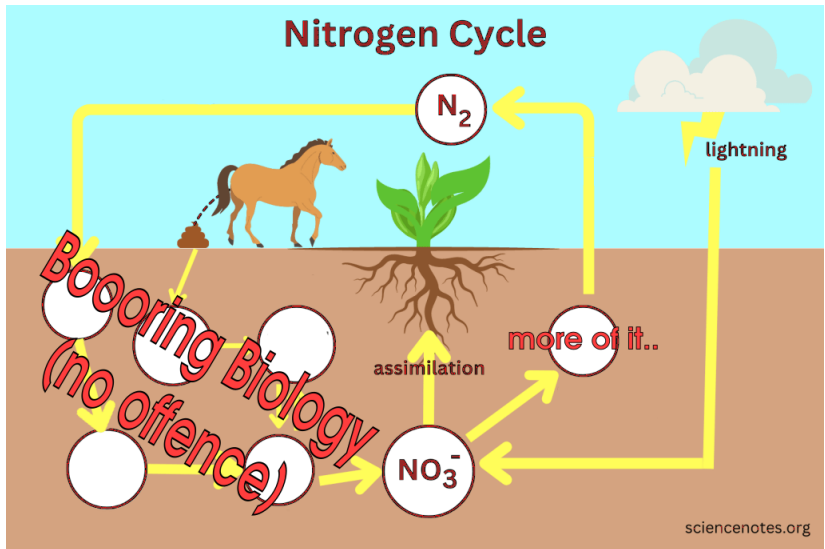
PFAS can be measured in **every third site**. The four highly regulated compounds¹ in 24%.

¹PFOS, PFOA, PFBS, PFHxS



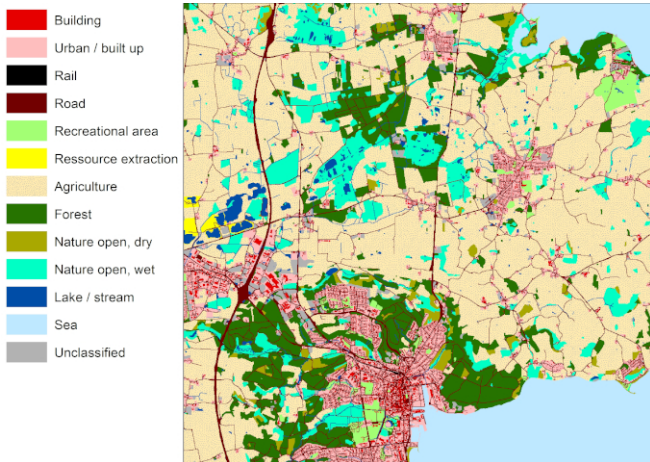


Nitrate - the Classic



Adapted from <https://sciencenotes.org/nitrogen-cycle/>





More than half of the area of Denmark is used for agriculture!²

²<https://www.dst.dk/en/Statistik/emner/miljoe-og-energi/areal/arealopgoerelser>

Of that only 4 % for direct human consumption³!

³<https://www.dyrenesbeskyttelse.dk/artikler/danmark-har-verdens-stoerste-koedindustri>

BREAKING: FARMERS TO BE SLIGHTLY LESS RICH!



Foto: TV2 ØJ

Landmænd raser over nyt lovforslag om kvælstofudledning

**MELLEMFOLKELIGT
SAMVIRKE** **actionaid**

BREAKING: FARMERS TO BE SLIGHTLY LESS RICH!



BREAKING: FARMERS TO BE SLIGHTLY LESS RICH!

Food waste in Denmark

Sumtotal 881.062 tons / year

Food industry 457,172 metric tons / year (51,9%)

Households 235,000 metric tons / year (26,7%)

Detail og engros 103,906 metric tons / year
(11,8%)

Primary production 42,984 metric tons / year (4,9
%)

Service sector 42,000 metric tons / year (4,8 %)

<https://stopspildafmad.org/om-madspild/madspild-i-tal/>

MEET THE MEAT THAT REALLY SHOULD MEET ITS END



[https://www.dyrenesbeskyttelse.dk/artikler/
danmark-har-verdens-stoerste-koedindustri](https://www.dyrenesbeskyttelse.dk/artikler/danmark-har-verdens-stoerste-koedindustri)

MEET THE MEAT THAT REALLY SHOULD MEET ITS END



or ~ 1.92 mio metric tons /
year. Exports $\sim 85\%$ to 90% ⁴

⁴[https://www.dyrenesbeskyttelse.dk/artikler/
danmark-har-verdens-stoerste-koedindustri](https://www.dyrenesbeskyttelse.dk/artikler/danmark-har-verdens-stoerste-koedindustri)

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CONSEQUENCES

Environmental Pollution of bodies of water. Excess algae growth, extinction of fauna, etc.

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Health Nitrate in drinking water has been linked to methemoglobinemia (blue-baby-syndrome), colorectal cancer, thyroid disease, and more [Ward et al., 2018, Jacobsen et al., 2024].
Increased risk even below regulatory limits [Ward et al., 2018]!

CONSEQUENCES FROM THE TAB

For over 60 years, the global guideline for nitrates in drinking water has been 50 milligrams per litre (mg/L) [...] set in the 1950s⁵.

⁵ <https://www.greenpeace.org/international/story/82511/industrial-agriculture-profits-toxic-nitrates-pollution-tap-water/>

This limit is hopelessly out of date

Literature proposes lower limits of $< 10\%$, reducing the risk by 15%. Even lower limits of $< 4\%$ bring this number down further [Jacobsen et al., 2024, Schullehner et al., 2018].

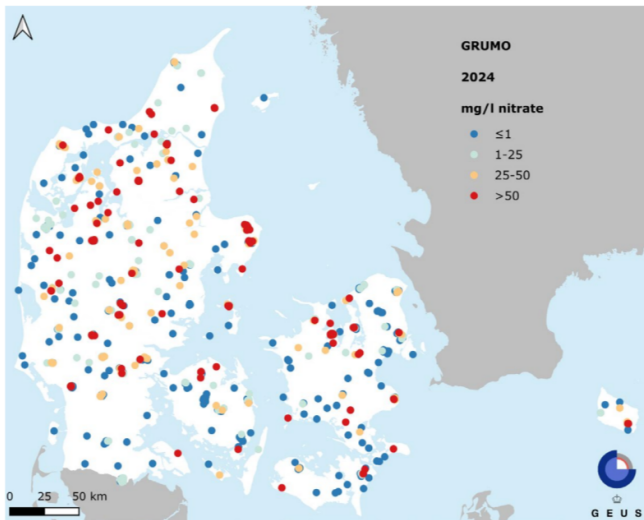
BLUE BABIES (NOT GOOD)



Infant methemoglobinemia

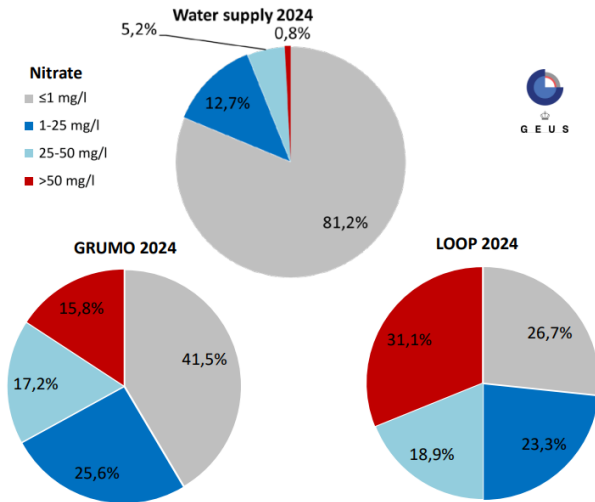
- ▶ Can progress to cause coma and death if not left untreated [Knobloch et al., 2000, Chen and Nappe, 2026].
- ▶ 2 cases where water for formula contained concentrations of **roughly half** of the 'safe limit' [Knobloch et al., 2000].

NITRATE POLLUTION IN DENMARK



From GEUS' groundwater report 2025 [Lærke Thorling et al., 2025].

MIXING AND EXPLOITING



From GEUS' groundwater report 2025 [Lærke Thorling et al., 2025].



PFAS - the Immortal

WHAT'S A PFAS?

Per- and poly**F**luoro**A**lkyl
Substances

WHAT THEY DO TO US⁶

- ▶ Decreased fertility, effects on pregnant people
- ▶ Developmental effects or delays in children,
- ▶ Risk of some cancers, prostate, kidney, testicular.
- ▶ Impact on immune system, infections, reduced vaccine response.
- ▶ Interference hormones.
- ▶ Increased cholesterol - risk of obesity.

⁶[https:](https://www.epa.gov/pfas/our-current-understanding-human-health-and-environmental-risks-pfas)

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WHERE DO THEY LIVE?

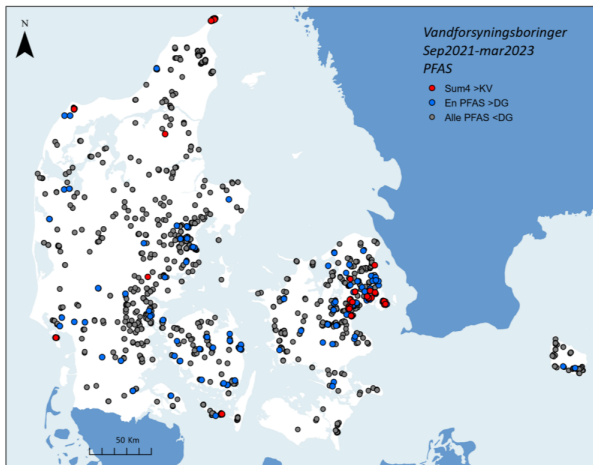
Where **DON'T** they live!? PFAS are found in water, air, fish, and soil..

WHERE DO THEY LIVE?

Where DON'T they live!? PFAS are found in water, air, fish, and soil..

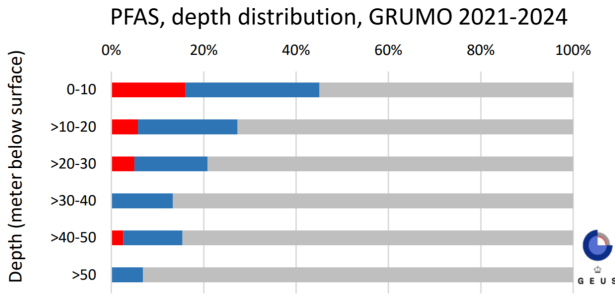
Also, in the blood of people and animals, and in food products⁷

PFAS MEASUREMENTS IN GROUNDWATER WELLS



Red: PFAS chemicals above limit. *Blue:* Measured PFAS chemicals. *Grey:* None detected [Anders R. Johnsen, Lærke Thorling & Christian N. Albers, 2023].

PFAS MEASUREMENTS IN GROUNDWATER WELLS



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PFAS does not simply break down. They're the eternal chemicals.

ToC

Part Two:

- ▶ Case studies in the global south
- ▶ How are we as Europeans involved?
- ▶ How is climate change changing the picture?

WATER IN KENYA



WATER IN KENYA

Article 42 of the Constitution of Kenya, 2010: “Every person has the right to a clean and healthy environment”, while Article 43 (d) of the Constitution guarantees the: “Right to clean and safe water in adequate quantities to every person.”

SOME NUMBERS [noa,]

- ▶ Only about 15 percent of Kenya's water resources are easily accessible, and the majority of that is surface water.

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- ▶ Still over-abstracted - water level decline and quality deteriorates.
- ▶ Poor legislation and mapping - 35.000 unregistered boreholes

WATER IN KENYA



Black water river running through the Kenyan slum next to Dandora landfill
©Christina Anderskov

WATER IN KENYA

Kenya generates between 3,000 to 4,000 tons of waste per day.

Nairobi generates between 2,000 to 2,500 tons of waste daily.

Less than half of that amount is re- or upcycled.

It is dumped in landfills, along roads, in streams or rivers, or burned.

This happens with little awareness of the subsurface and thus groundwater.⁸

⁸<https://eng.mst.dk/about-the-danish-epa/global-cooperation/kenyacircular-economy>

WATER IN ETHIOPIA



WATER IN ETHIOPIA⁹

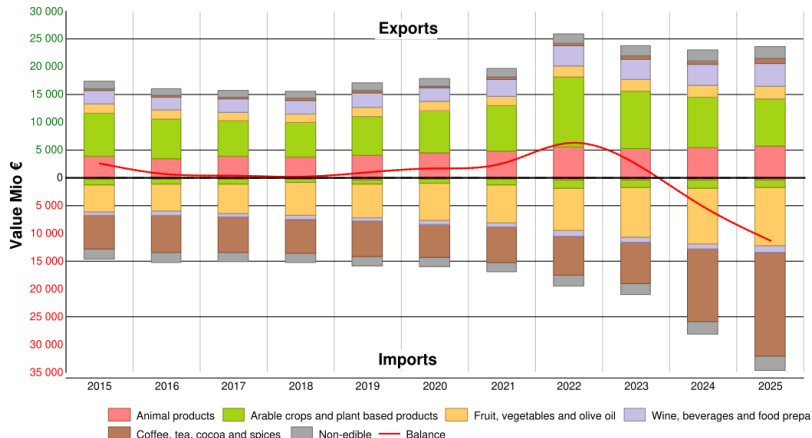
- ▶ 13% have access to safe drinking water - most of it from the ground.

Particularly impacts the rural population.

Vast, but unevenly distributed, resources of Groundwater.

⁹ <https://eng.mst.dk/about-the-danish-epa/global-cooperation/ethiopia>

EU TRADE WITH AFRICA



Import from and exports to Africa¹⁰

¹⁰Data from https://agriculture.ec.europa.eu/system/files/2023-05/agrifood-africa-all-countries_en.pdf

TRADE WITH AFRICA

- ▶ Vast mineral exploitation by Western powers - Kabanga Nickel Project. No African representatives¹¹.
- ▶ Massive farming subsidies in the EU/west outcompete local producers [*Arenas and Grobusch, 2021*].

¹¹ <https://www.cetri.be/Foreign-countries-are-lining-up-to-6536?lang=fr>



Climate - It's coming for you



Flood move a great amount of pollutants into water resources
and moves natural barriers.

2025 GREECE

Kort nyt | 3. jul 2025

Græsk ferieø hærget af skovbrand - igen



Omkring 3.000 beboere og turister er blevet evakueret. (Foto: © STEFANOS RAPANIS, Ritzau Scanpix)

Fires require access to large amounts of water for controlling and preventing them.

CONGO 2026

Jordskred ved mine i DR Congo dræber over 200 mennesker - mange af dem børn



Billedet her af Rubaya-minen i det østlige DR Congo er fra den 30. januar, hvor minen ligeledes blev ramt af et jordskred. Minen kontrolleres af oprørsgruppen M23 og står for en betydelig del af verdens produktion af coltan, der bruges i elektronik som mobiltelefoner og computere. (Foto: © AFP, Ritzau Scanpix)

Causes destruction and pollution of water sources and infrastructure

Particularly affects vulnerable populations in the Global South *[Ngcamu, 2023]*

Particularly affects vulnerable populations in the Global South *[Ngcamu, 2023]*

- ▶ Increases in droughts and extreme weather
- ▶ Unable to protect themselves
- ▶ Systematically overlooked

A shirtless male boxer is shown in a boxing ring, with his face obscured by a white circle containing the text 'Us'. He is in a fighting stance with his fists raised. The background shows the yellow ropes of the ring and a blurred crowd of spectators.

Us

WHAT DO?

Protect groundwater resources by protecting 160,000 hectares
30% Nature Goal EU biodiversity rules.
Farm Reforms by slashing emissions from pig farming.

THE REALITY CHECK:

- ▶ Backlash and intense lobbying from Landbrug & Fødevarer¹² Wanting to reopen negotiations.

¹²<https://nyheder.tv2.dk/live/politik/2026-02-26-folketingsvalg-2026/groen-trepart-kan-vaere-i-fare-advarer-landbrug---foedevarer?entry=1b3e7e9c-5079-46c0-ade9-20fdacf8161b>

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Denmark still is one of the most cultivated countries. This will guide political conservatism.

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WHAT DO?

Buy organic to avoid pesticides.

Buy local and in season to avoid water and energy hungry greenhouses and imports.

Reduce meat and dairy as a massive fraction of large scale farming is for fodder.

Ditch Teflon for steel, cast iron - these are fine and good. They last. Find them second hand.

Buy non-coated clothes and avoid chemical waterproofing.

Reuse / recycle

Stay informed!!!

CONSUMER POWER (HEY, THAT'S US!)

Stay informed and check out: PFAS free products



Or: Shopper's Guide to Avoiding PFAS



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